DDM ClickHouse vs PostgreSQL Queries

Query 1

Show all reports where the building type is a Private Home OR Villa, that happened on a weekend (Day of Week = 6 or 7), AND where the call duration was above 30 minutes. Order the results by call duration descending.

SQL

```
SELECT *
FROM noisy_neighbors
WHERE building_type IN ('Private Home', 'Villa')
  AND day_of_week IN (6, 7)
  AND duration_of_call_min > 30
ORDER BY duration_of_call_min DESC;
```

Query 2

Within each zip group and day type, keep only the 2 longest calls, breaking ties by earlier report date. Where 'North' means zip 10000-10499, 'Center' means zip 10500-10999, 'South' means zip 11000-11499, 'Weekday' means day of week 1-5, and 'Weekend' means day of week 6 or 7.

SQL

```
WITH classified AS (
    SELECT
        *,
        CASE
            WHEN incident_zip BETWEEN 10000 AND 10499 THEN 'North'
            WHEN incident_zip BETWEEN 10500 AND 10999 THEN 'Center'
            WHEN incident_zip BETWEEN 11000 AND 11499 THEN 'South'
        END AS zip_group,
        CASE
            WHEN day_of_week BETWEEN 1 AND 5 THEN 'Weekday'
            ELSE 'Weekend'
        END AS day_type
    FROM noisy_neighbors
    WHERE incident_zip BETWEEN 10000 AND 11499
      AND day_of_week BETWEEN 1 AND 7
),

ranked AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY zip_group, day_type
            ORDER BY
                duration_of_call_min DESC,
                date_of_report ASC
        ) AS rn
    FROM classified
)

SELECT *
FROM ranked
WHERE rn <= 2
ORDER BY
    zip_group,
    day_type,
    duration_of_call_min DESC;
```

Query 3

For each month, compute a monthly metric based on call durations, active report days, and the most frequent borough.

METRIC DEFINITION

$$\text{metric} = \left(\frac{\sum \text{Duration of Call (min)}}{\text{unique report dates in the month}} \right) \times \text{length of the most frequent borough name in that month}$$

SQL

```
WITH base AS (
    SELECT
        *,
        TO_DATE(date_of_report) AS report_date,
        DATE_FORMAT(TO_DATE(date_of_report), 'yyyy-MM') AS year_month
    FROM noisy_neighbors
),
monthly_stats AS (
    SELECT
        year_month,
        SUM(duration_of_call_min) AS sum_duration,
        COUNT(DISTINCT report_date) AS uniq_dates
    FROM base
    GROUP BY year_month
),
borough_counts AS (
    SELECT
        year_month,
        borough,
        COUNT(*) AS borough_reports
    FROM base
    GROUP BY year_month, borough
),
max_per_month AS (
    SELECT
        year_month,
        MAX(borough_reports) AS max_borough_reports
    FROM borough_counts
    GROUP BY year_month
),
top_borough_candidates AS (
    SELECT
        bc.year_month,
        bc.borough,
        bc.borough_reports
    FROM borough_counts bc
    INNER JOIN max_per_month mp
        ON bc.year_month = mp.year_month
    WHERE bc.borough_reports = mp.max_borough_reports
),
top_borough AS (
    SELECT
        year_month,
        MIN(borough) AS top_borough,
        LENGTH(MIN(borough)) AS top_borough_len
    FROM top_borough_candidates
    GROUP BY year_month
)
SELECT
    ms.year_month,
    ms.sum_duration,
    ms.uniq_dates,
    tb.top_borough,
    (ms.sum_duration / ms.uniq_dates) * tb.top_borough_len AS monthly_metric
FROM monthly_stats ms
INNER JOIN top_borough tb
    ON ms.year_month = tb.year_month
ORDER BY ms.year_month ASC;
```